

Additional Transformation to Eve Online Market Dataset

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When you're getting a dataset ready for solid analysis, the first step is to tackle missing values in a straightforward but thoughtful way. I would start by checking which columns have gaps, then decide whether those blanks really mean a value is "unknown" or if they're just oversights in general. For numeric fields, we might fill in with the median value or even develop some sort of predictive model to guess the missing bits. For category fields, sometimes it makes sense to add a "Not Specified" label or swap in the most common entry in the dataset.

Next up is outlier detection, because weird spikes or dips can really throw off averages and models, especially when dealing with numerically heavy datasets. We can use interquartile ranges or z-scores to spot those unexpected values, then decide if we cap them, transform them, or remove them entirely. The key is not to erase something important, but to make sure one extreme order of a million units in the market doesn't warp our insights about typical customer behavior.

It's also vital to enforce the correct data types across the board. Dates should be real datetime objects, quantities as integers, and prices as floats. This step might seem trivial and not important, but it keeps the Pandas library from throwing obscure errors and lets us run date math or numeric summaries without complex calculations. Removing duplicate records is another big one. For example, we can scan for repeated order IDs or records that match on key fields, then drop the extras. That way, our counts and summary stats actually reflect the unique orders and not multiples of the same order, which may throw off our analysis.

If there are different units (some weights in pounds and others in kilograms, or prices in various currencies), we'll standardize everything to one unit, so comparisons are standardized. If

there are any free-text fields like comments or descriptions, we can standardize and normalize them by lowercasing, stripping punctuation, and maybe doing basic tokenization. This preps the text for any analysis or keyword extraction we might want later.

Finally, we can add extra value with some new features that can boost our analysis, like calculating lead times by subtracting the issue date from the last modified date, pulling out weekday or month flags, or creating rolling counts of past orders. These extras often unlock insights that a raw table alone can't provide and lead us to new questions, which in turn lead us to more transformations, analysis, and questions until the cycle is no longer necessary or needed.